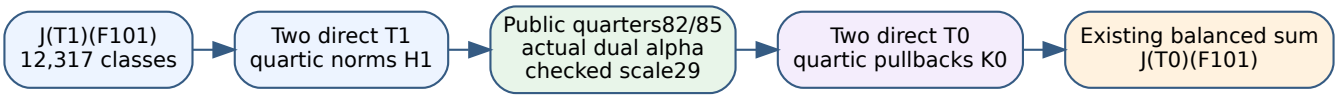
# Prime-field adjoint and the next two quartic covers

2026-11-12 · fixed F101 models · scientific standing unsigned

The first Jacobian transfer now has a direct prime-field adjoint prototype. Native24/PCCdcbb58 checks every12,317 concrete olddual output, bijection, all fixed-partner sums, negations, identity and wrong-model inputs. Both compositions with the accepted prime-field forward evaluator equal multiplication by7 on every class in both rational domains. Every109/113 reverse embedded elliptic-source endpoint is also checked. StrictClippy passes. These are local execution results with unsigned standing. [pcc:unverified]




## Exact reverse maps and correctness argument

The reverse elliptic kernels are[58,26,58,1] on(26,58) and[29,88,45,1] on(69,50). Existing Kohel produces codomains(78,72)/(99,68);the checked scale29=1/7 in F101 returns(1,46)/(44,21). Kernel, numerator, denominator, domain, codomain and scale match the retained old maps. Constructor and evaluation call existing F101 norm, polynomial, elliptic, Kohel and balanced engines. Old extension-field reference and class enumeration are separate. [pcc:unverified]

Taking adjoints of4Psi=K1 alpha H0 gives4PsiHat=K0 alphaHat H1. On the rational groups of order12,317=109·113, the component inverse4 scalars82/85 yield the old adjoint. This is a rational-point formula; the old global norm, Picard and polarization identities remain explicit dependencies. PRIME_FIELD_ADJOINT_PROOF.md/PCC526f9f records this argument. No global inverse morphism of multiplication by4 is asserted. [pcc:unverified]

| Component | Retained evidence |
|---|---|
| T1 norms | c3a1a7:both12,317 domains,4linear/232partial-pole inputs each |
| T0 pullbacks | 5143f5:oldR0P0,norm4,group/infinity controls |
| Whole adjoint | dcbb58:olddual equality and both full norm7 compositions |
| Written geometry | 4934b4/78176c/526f9f;partial formalization remains |

## Direct source control and preserved corrections

Source control22/839704 checks every109/113 input/scalar and the exact endpoint PsiK0=K1alpha, target4alpha, reverse28, injectivity and wrong-quarter rejection. The correct inverse28 is74 modulo109 and109 modulo113. SOURCE_ENDPOINT_CONTROL_V2.md/c2b7c9 corrects the frozen d55462 write-up; failed native21/122fda remains. Cache19 selectedzero tests and was rejected; isolated20 and adjoint23 lint failures also remain. Each scratch package now has its own Cargo target. No failed attempt is promoted to a pass. [pcc:unverified]

# T2: exact quartics and a different pole splitting field

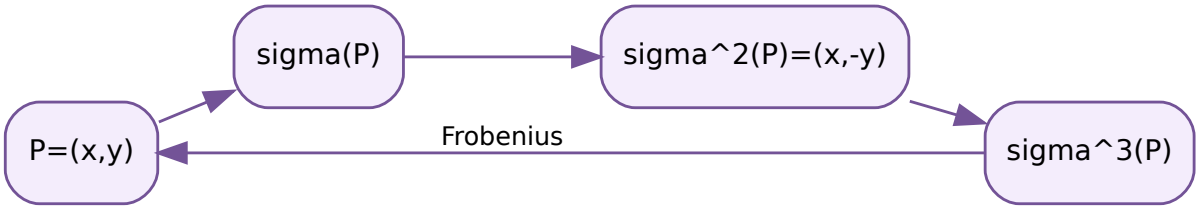
T2:$y^2=F2(x)$, F2=[95,93,43,39,92,17,1]. The maps have $X=N/Q^2$, $Y=yM/Q^3$, with low-to-high coefficient lists:

| | |
|---|---|
| E109 double-prime=(59,27) | E113 double-prime=(23,7) |
| N=[97,93,7,0,97] | N=[74,0,81,42,64] |
| Q=[55,0,1] | Q=[23,0,1] |
| M=[71,49,20,63] | M=[37,9,9,35] |
| infinity+=(97,63) | infinity+=(64,35) |
| 118finite witnesses;0rational poles | 116finite+2origin witnesses |

Construction25/412aba derives these candidates from the exact oldH2=N2R2Hat outputs. The original three T0/T1 constructor tests pass after sharing the same reconstruction body. Independent flat-field28/2fee3f checks exact curve identities, source irreducibility, smoothness, coprimality, separable degree4, complete118-point finite domains, both infinities, all four simple geometric poles and wrong-codomain controls. Full changed standalone29 passes all19 tests with original guards and strictClippy, PCCbfd777. [pcc:unverified]

## A retained negative result about the pole field

The first T2 cover has all four pole points in one Frobenius orbit of length4. In $F101[z]/(z^4-2)$, the checker verifies x not inF101, $x^{(101^2)}=x$ and $y^{(101^2)}=-y$, with fourth Frobenius fixing each point. These points are absent from$F101^6$. The failed degree-six assumption27/ bf2591 is retained. The second cover has rational poles(49,12)/(49,89) and its complete pole fibre is checked in the declared$F101^6$ representation. [pcc:unverified]




T2_QUARTIC_PROOF.md/f9cf19 identifies both covers with the old centred maps:the old norm/Picard identity gives degree4, Milne's arbitrary-map cube bounds a difference by16, and116/118 distinct agreements exceed that bound. The same finite-algebra pullback and infinity bookkeeping applies. Pole splitting fields do not change the F101 coefficients and do not require choosing pole roots in the prime-field evaluator. [pcc:unverified]

## Current integration and next acceptance

The actual whole-forward consumer check.sh04 is running on unchanged eight source02 files after the workspace restart interrupted03;partial03/source custody f87a99/root2fe5dc is retained. Expected25native/90reference+five integration tests is pending. Parent1505 exact6cf focusedCI43165729371 passes25native/85reference+five, fullfd53e1/root282f74. Root33e exactCI a990e6 covers17standalone/136written helpers/25modules;the current19-test scope needs its source-changing checkpoint and ownCI. [pcc:unverified]

T2 directnorm/pullback adapters and a prime-field second hop are next. Matched cold costs await S3/profile/livebudget admission. KB/ecdb remain unavailable and the autoresearcher atlas remains vault locked. Sources: the named written proofs, Milne AV Cor5.3 and Galbraith Chapters8/10. No timing or novelty claim is made. [pcc:unverified]